Our country, our future

525/1

S6 CHEMISTRY

Exam 24

PAPER 1

DURATION: 2 HOUR 45 MINUTES

Instructions:

1. Answer all questions in the spaces provided
2. Illustrate your answers with equations wherever possible
3. A graph paper and periodic table are attached at the end of the paper.

1. Complete the following equations and give the IUPAC name of the main organic product

(a) $CH_3CH_2Br + HC \equiv \bar{C}Na^+ \xrightarrow{liq.NH_3}$ (2m)

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(b) $nCH_3CH = CH_2 \xrightarrow{catalyst.}$ (2m)

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2. Element X, Y and Z have the following ionisation energies ($kjmol^{-1}$)

Ionisation energy	1st	2nd	3rd	4th
X	738	1450	7730	10550
Y	495	4653	6912	9540
z	800	2427	3658	25024

Select the element that shows

(i) a valency of +2 (½ m)

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(ii) forms a covalent chloride (½ m)

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(iii) forms an ionic univalent chloride (1 m)

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(b) Explain why the first ionisation energy of Boron is less that that of Beryllium, whereas the second ionisation energy of Boron is greater than that of Beryllium (3m)

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3. (a) State Raoult's law (2)

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(b) The vapour pressure of a solution containing 4g of a sugar Q in 100g of water is 3.1545 Kpa at 25°C. Calculate the relative molecular mass of Q given that the vapour pressure of pure water at 25°C is 3.167 Kpa. (3m)

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4. The molecular formula of a hydrocarbon X is C_4H_8

(a) Write the IUPAC names and structural formula of all possible isomers of x (3)

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(b) Ozonolysis of x followed by hydrolysis gave only one compound. Identify the compound (1m)
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(c) Write equation to show how x can be converted to alkyne. (2m)
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5. (a) Define the term partition coefficient (2m)
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(b) 200cm^3 of an aqueous solution contains 8gm of Y. Calculate the mass of y left in the aqueous layer by shaking the solution with

(i) 50cm^3 of trichloromethane. (The partition coefficient of Y between trichloromethane and water is 12) (4m)
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(ii) twice with 25cm^3 of trichloromethane (4m)

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6. (a) (i) State what is meant by an azeotropic mixture (2m)

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(ii) Give two reasons why an azeotropic mixture is not considered a pure substance (1m)

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(b) ethanol and water forms an azeotropic mixture. Explain (3m)

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7. 2 -Bromo-2-methyl propane reacts with aqueous sodium hydroxide.

(a) (i) Write an equation for the reaction (1m)

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(ii) Write the rate equation for the reaction (1m)

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(iii) State the molecularity of the reaction (1m)

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(b) Draw the energy diagram for the reaction (3m)

8. (a) State Hess' law (2m)

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(b) State four factors that affect the enthalpy change of any system.

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(2m)

(c) The enthalphy of combustion of ethene is – 14411 $kJmol^{-1}$ while that of ethyne is – 1299.5 $kJmol^{-1}$ and that of hydrogen is – 285.8 $kJmol^{-1}$ Calculate the enthalpy change of hydrogenation of ethyne to ethene.

(4 m)

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9. Complete the following equations and in each case write a mechanism for the reaction

(a) —CH_2I + $\bar{O}H_{(aq)}$ $\xrightarrow{\text{heat}}$ (3m)

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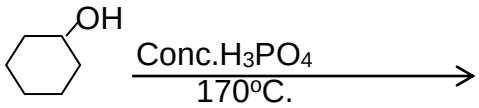
(b) 

(3m)

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10. (a) Explain what is meant by steam distillation. (Diagram not required (3m)

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(b) Calculate the percentage composition by mass of water when aniline ($C_6H_5NH_2$) is steam distilled at 760mm Hg and 98°C and the vapour pressure of water at 98°C is 734 mmHg. (3m)

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11. 0.80g of a metal oxide of the type MO was dissolved in 50.0cm^3 of 1M hydrochloric acid. The resulting mixture was made to 250cm^3 with distilled water. 25.0cm^3 of this solution required 21.4cm^3 of 0.1M sodium hydroxide solution for neutralisation.
Calculate the
(a) the number of moles of excess hydrochloric acid in 250cm^3 (3m)

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(b) (i) The mass of oxide MO that reacted with the acid (2m)

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(ii) formula mass of MO (1m)

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(iii) the relative atomic mass of M (1m)

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12. Give reaction scheme by which the following conversions can be effected

(a) Methane from chloroethane (3m)

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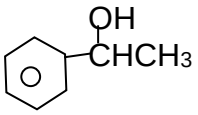
(b) 

from (3m)

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13. (a) Fill in the blanks in the following radioactive decay series

(i) $^{235}U \xrightarrow{\alpha}$ ----- $\xrightarrow{\beta}$ ----- $\xrightarrow{\alpha}$ ^{227}AC

(2m)

(ii) ---- $\xrightarrow{\alpha}$ ^{233}Pa $\xrightarrow{\beta}$ ------ $\xrightarrow{\alpha}$ (2m)

(b) A radioactive substance decays as follows:

Time (day)	0	1	2	3	4	5	6
Mass (g)	500	389	303	236	184	143	112

(i) Plot a graph of time against mass

(ii) Determine the decay constant and half life of the reaction

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END

Good luck